

1 Verilog Implementation

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1 module phase3(  
2     input  clk,  
3     input  rst,  
4     input  [2:0] code_in,  
5     output reg phase3_done,  
6     output reg phase3_fail  
7 );  
8 parameter safe=3'b000;  
9 parameter up=3'b001;  
10 parameter down=3'b010;  
11 parameter right=3'b011;  
12 parameter left=3'b100;  
13 parameter stay=3'b101;  
14 parameter done=3'b110;  
15 parameter fail=3'b111;  
16 reg alaram;  
17 reg [2:0] current_state,next_state;  
18 always @(posedge clk) begin  
19     if(rst)  
20         current_state<=safe;  
21     else  
22         current_state<=next_state;  
23 end  
24 always @(*) begin  
25     phase3_done=0;  
26     phase3_fail=0;  
27     alaram=0;  
28     case(current_state)  
29         safe:  
30             next_state=up;  
31         up:  
32             if(code_in==3'b000)  
33                 next_state=down;  
34             else  
35                 next_state=fail;  
36         down:  
37             if(code_in==3'b001)  
38                 next_state=left;  
39             else  
40                 next_state=fail;  
41         left:  
42             if(code_in==3'b010)  
43                 next_state=right;  
44             else  
45                 next_state=fail;  
46         right:  
47             if(code_in==3'b011)  
48                 next_state=stay;  
49             else
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50         next_state=fail;
51     stay:
52         if(code_in==3'b100)
53             next_state=done;
54         else
55             next_state=fail;
56     done:
57         phase3_done=1;
58     fail: begin
59         phase3_fail=1;
60         alaram=1;
61     end
62     default:
63         next_state=safe;
64 endcase
65 end
66 endmodule

```